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Ensemble Surrogate Archive-Assisted Artificial Bee Colony–NSGA-II Metaheuristics for Multi-Modal Optimal Power Flow Framework

Abhishek Katkar
Department of Electrical Engineering
Government Polytechnic,
Kolhapur, Maharashtra, India
ORCID: 0009-0009-0045-4786

Pravin Jadhav
Department of Humanities and Social Sciences
IITRAM,
Ahmedabad, Gujarat, India
Email: pravinjadhav@iitram.ac.in

Abhijeet Redekar
Department of Electrical Engineering
BSIET,
Kolhapur, Maharashtra, India
Email: abhijeet.redekar@gmail.com

Himmat Jadhav
Faculty of Science and Technology
S.N.D.T. Women's University,
Mumbai, Maharashtra, India
Email: htj.rit@gmail.com

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***Abstract*—Addressing multimodal multi-objective optimization problems (MMOPs) presents a considerable challenge, primarily due to the presence of conflicting objectives and the necessity for convergence and diversity within both the Pareto Front (PF) and decision space. A novel framework, termed *Ensemble Surrogate Metamodel Archive* (ESMA), is proposed to enhance surrogate-based approximation accuracy by integrating Radial Basis Function (RBF), Kriging (Gaussian Process Regression), and Support Vector Regression (SVR) models. The ESMA employs a hybrid Multi-Objective Artificial Bee Colony and NSGA-II algorithm (MOABC–NSGA-II) to effectively manage multiple non-dominated Pareto optimal solutions in the context of constrained multi-modal optimal power flow problems. The framework includes support for external archives and is evaluated against two established archiving strategies: Dynamic Reference Space-based Clustering (DRSC) and Pareto-Crowded Sorting Dominance (PCSD). Validation is conducted on the IEEE 30-bus system within the framework of CMCMMOPF conditions. The results indicate that ESMA–MOABC–NSGA-II exhibits superior convergence speed, improved diversity preservation, and produces high-quality solutions. The method achieves an average success rate of 88.69% with a computation time of 3.01 seconds, markedly surpassing DRSC–MOAGDE (66.87%, 9.47 s) and PCSD–IMOMRFO (64.29%, 1372.49 s). In six real-time scenarios, the proposed framework demonstrates average reductions of 0.86% in total fuel cost, 4.31% in emissions, 30.29% in voltage deviation, and 49.19% in active power loss, outperforming current state-of-the-art methods. The results validate the effectiveness and reliability of the proposed framework in advanced optimization computation.**

***Index Terms*—Constrained Multi-objective optimization, Ensemble surrogate metamodel, Artificial Bee Colony, NSGA-II, Optimal Power Flow, Dynamic reference space clustering , Pareto-crowded sorting dominance,**

I. INTRODUCTION

Modern power systems are becoming more complex due to renewable energy integration, changing load patterns, and strict operational limits, requiring effective Multi-Objective Optimal Power Flow solutions. CMMOPF seeks to reduce conflicting goals like generation cost, emissions, and losses, while following nonlinear equality and inequality constraints. Multiple global or local Pareto-optimal regions define the problem as a Multimodal Multi-Objective Optimization Problem (MMOP), complicating convergence and diversity preservation [2]. Traditional algorithms like NSGA-II and MOABC

TABLE I
LIST OF NOTATIONS AND ABBREVIATIONS

Parameter	Description
MOPF	Multi-Objective Optimal Power Flow
CMMOP	Constrained Multimodal Multi-Objective Problem
ABC	Artificial Bee Colony
NSGA-II	Non-Dominated Sorting Genetic Algorithm II
MOABC	Multi-Objective Artificial Bee Colony
ESMA	Ensemble Surrogate Metamodel Archive
RBF	Radial Basis Function
SVR	Support Vector Regression
DRSC	Dynamic Reference Space-based Clustering
PCSD	Pareto-Crowded Sorting Dominance
X_i	i-th candidate solution vector
D	Number of decision variables
N	Population size
$G_{\max}$	Maximum number of generations
$f_i(x)$	i-th objective function
$h_k(x)$	k-th equality constraint
$g_j(x)$	j-th inequality constraint
$\tilde{f}(x)$	Penalized fitness function
$CD(i)$	Crowding distance of individual i
ϕ_n	Crowded-comparison operator in NSGA-II
T_{update}	ESMA retraining interval
$P_{\text{THG}i}$	Active power output of thermal generator i
$Q_{\text{THG}i}$	Reactive power output of thermal generator i
$V_{\text{THG}i}$	Voltage magnitude at generator bus i
T_j	Tap setting of transformer j
Q_{Ck}	Reactive power of shunt compensator at bus k

often face issues with premature convergence and limited exploration in these landscapes. They typically fail to maintain diverse Pareto-optimal areas and have difficulty managing redundancy in the decision space [3]. Static archive management strategies lack adaptability, limiting their ability to balance exploration and exploitation in complex multi-modal power system optimization scenarios [4].

Surrogate-assisted evolutionary algorithms (SAEAs) offer efficient solutions to high computational costs and premature convergence in multimodal multi-objective problems. Surrogate models like Radial Basis Function (RBF), Kriging, and Support Vector Regression (SVR) often struggle to adapt in high-dimensional and nonlinear search spaces [5]. We propose an Ensemble Surrogate Metamodel Archive (ESMA) that integrates RBF, Kriging, and SVR to enhance prediction accuracy and support exploration of different fitness landscapes [6]. This ensemble model is part of a new hybrid optimization framework, ESMA–MOABC–NSGA-II, aimed at effectively tackling constrained Multi-Modal Optimal Power Flow (CMMOPF) problems [6]. The proposed framework is validated with the IEEE 30-bus system and compared to leading methods, demonstrating improved convergence, accuracy, and robustness in complex power system scenarios [7].

The ESMA–MOABC–NSGA-II framework, illustrated in Fig. 1, is assessed alongside two advanced archive-assisted algorithms: the Dynamic Reference Space-based Clustering (DRSC) mechanism with MOAGDE, and the Pareto-Crowded Sorting Dominance (PCSD) strategy paired with IMOMRFO [8]. Results show that ESMA–MOABC–NSGA-II achieves better convergence and a more balanced Pareto front [9]. Advancements come from a cooperative hybrid engine that combines MOABC's global exploration with NSGA-II's elite sorting and diversity preservation methods [9]. Experimental validations show the framework's effectiveness in convergence speed, diversity retention, solution quality, and computational efficiency under constrained CMMOPF conditions. Table I displays the notations and abbreviations used in this research.

II. Archiving Strategies in Multi-objective Optimization

Archiving strategies in multi-objective optimization are essential for preserving non dominated solutions and ensuring diversity along the Pareto front during the evolutionary process. [10] Effective archiving mechanisms, including clustering-based, preference-guided, and surrogate-assisted methods, facilitate a balance between convergence and diversity, particularly in complex and constrained search spaces.

A. Ensemble Surrogate Metamodel Archive (ESMA)

The proposed framework addresses the computational challenges of MMOPs by using an ESMA that integrates Radial Basis Functions (RBF), Kriging (Gaussian Process Regression), and Support Vector Regression (SVR) into a unified prediction model. [10] Each surrogate has unique benefits:

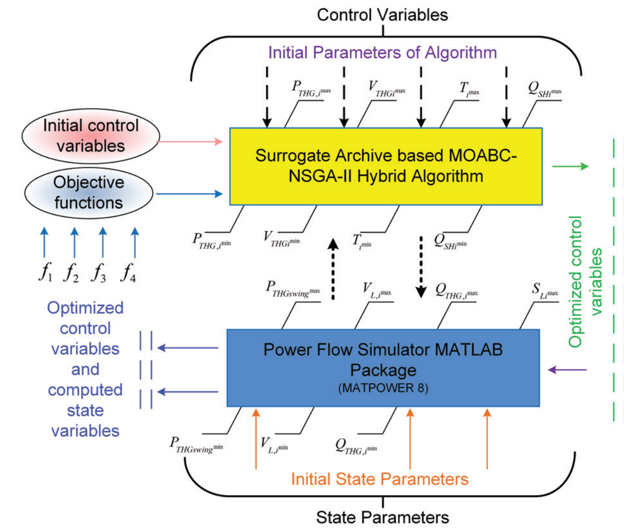

Fig. 1. ESMA-CMMOPF Architecture using MOABC-NSGAII

RBF offers smooth local interpolation, Kriging accounts for uncertainty in estimates, and SVR enhances generalization in high-dimensional, sparse settings.

Let $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{f}_i)\}_{i=1}^{N}$ represent the dataset of N evaluated solutions, where $\mathbf{x}_i \in \mathbb{R}^d$ is the decision vector and $\mathbf{f}_i \in \mathbb{R}^m$ is the objective vector. The prediction of the ensemble surrogate at a new input $\mathbf{x}^*$ is expressed as:

$$\hat{\mathbf{f}}(\mathbf{x}^*) = \sum_{k=1}^{K} w_k \cdot \hat{\mathbf{f}}^{(k)}(\mathbf{x}^*), \quad \sum_{k=1}^{K} w_k = 1, \; w_k \geq 0, \tag{1}$$

where $\hat{\mathbf{f}}^{(k)}$ represents the prediction from the k-th surrogate model, with w_k as its associated weight, which is adaptively adjusted according to prediction error or the width of the confidence interval. This ensemble prediction evaluates candidate solutions and updates the adaptive archive of non-dominated solutions.

B. Dynamic Reference Space-based Clustering (DRSC)

The Archive management through a DRSC mechanism to maintain diversity in both decision space and objective space. [11] In contrast to static clustering, DRSC dynamically transitions among three reference spaces—$\mathcal{D}$ (DS), $\mathcal{F}$ (OS), and the unified space $\mathcal{U} = \mathcal{D} \cup \mathcal{F}$—during archive reduction.

Let $A = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ be the archive of non-dominated solutions. A clustering process is applied in the dynamically selected reference space using K-means clustering. Given K clusters, the centroids $\{\mathbf{m}_1, \ldots, \mathbf{m}_K\}$ are determined and the closest solution $\mathbf{z}_i$ to each centroid is retained:

$$\mathbf{z}_i = \arg\min_{\mathbf{x} \in C_i} \|\mathbf{x} - \mathbf{m}_i\|^2, \quad i = 1, \ldots, K, \tag{2}$$

Let C_i represent the i-th cluster in the reference space, and let $\|\cdot\|$ denote the Euclidean norm. This method guarantees equitable representation throughout all regions of the PF and PS, while efficiently removing redundancy.

C. Pareto-Crowded Sorting Dominance (PCSD)

The PCSD method is utilized to enhance the archive's refinement. PCSD improves elitism and diversity through the integration of Pareto dominance and local crowding distance-based rankings [12]. For two non-dominated solutions $\mathbf{x}_i$ and $\mathbf{x}_j$, PCSD establishes a dominance preference based on:

$$\mathbf{x}_i \prec_{\text{PCSD}} \mathbf{x}_j \iff \begin{cases} \mathbf{x}_i \prec \mathbf{x}_j, & \text{(Pareto dominance)} \\ \text{or} \\ \text{rank}(\mathbf{x}_i) < \text{rank}(\mathbf{x}_j), & \text{(Crowding preference)} \end{cases} \tag{3}$$

This hybrid dominance strategy enables the algorithm to prioritize solutions that are both non-dominated and situated in sparsely populated areas of the search space, thereby fostering a well-distributed and diverse Pareto front.

III. Mathematical Formulation of the CMMOPF Problem [13]

The multi-objective power flow optimization problem entails identifying optimal control variables to concurrently minimize conflicting objectives, including fuel cost, emissions, voltage deviation, and power loss, while adhering to power system operational constraints as shown in Fig. 2. The formulation is as follows:

$$\begin{aligned} &\text{Minimize:} \quad \mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})] \\ &\text{Subject to:} \quad h_k(\mathbf{x}) = 0 \quad (k = 1, \ldots, K) \\ &\qquad g_j(\mathbf{x}) \leq 0 \quad (j = 1, \ldots, J) \end{aligned} \tag{4}$$

Objective functions:

1) Fuel Cost Minimization:

$$f_1 = \sum_{i=1}^{N_{THG}} (a_i + b_i P_i + c_i P_i^2) \tag{5}$$

2) Emission Reduction:

$$f_2 = \sum_{i=1}^{N_{THG}} (\sigma_i + \beta_i P_i + \tau_i P_i^2 + \omega_i e^{\mu_i P_i}) \tag{6}$$

3) Voltage Deviation Minimization:

$$f_3 = \sum_{n=1}^{NPQ} |V_n - 1| \tag{7}$$

4) Active Power Loss Minimization:

$$f_4 = \sum_{(i,j) \in NTL} G_{ij}(V_i^2 + V_j^2 - 2V_i V_j \cos\theta_{ij}) \tag{8}$$

Equality constraints (Power balance):

$$\begin{aligned} &P_i - P_{L_i} - V_i \sum_{j=1}^{N_{bus}} V_j (G_{ij}\cos\delta_{ij} + B_{ij}\sin\delta_{ij}) = 0 \\ &Q_i + Q_{SH_i} - Q_{L_i} - V_i \sum_{j=1}^{N_{bus}} V_j (G_{ij}\sin\delta_{ij} - B_{ij}\cos\delta_{ij}) = 0 \end{aligned} \tag{9}$$

TABLE II
Test Case Scenarios and Considered Objectives for IEEE 30-Bus System

Case	Fuel Cost (f_1)	Emission (f_2)	Voltage Deviation (f_3)	Power Loss (f_4)
Case 1	✓	✓	–	–
Case 2	✓	–	✓	–
Case 3	✓	✓	–	✓
Case 4	✓	✓	✓	–
Case 5	✓	✓	✓	✓
Case 6	✓	✓	✓	✓

Inequality constraints:

$$\begin{aligned} &P_i^{\min} \leq P_i \leq P_i^{\max}, \quad Q_i^{\min} \leq Q_i \leq Q_i^{\max} \\ &V_i^{\min} \leq V_i \leq V_i^{\max}, \quad T_i^{\min} \leq T_i \leq T_i^{\max} \\ &Q_{SH_i}^{\min} \leq Q_{SH_i} \leq Q_{SH_i}^{\max}, \quad S_{ij} \leq S_{ij}^{\max} \end{aligned} \tag{10}$$

Here, P_i, Q_i, V_i, and T_i are control variables; S_{ij} is the apparent power on line (i, j); and Q_{SH_i} is pertains to shunt VAR compensation. This formulation serves as the foundation for the application of surrogate-assisted hybrid metaheuristic algorithms in addressing constrained, multimodal CMMOPF problems.

IV. Hybrid MOABC–NSGA-II Algorithm: Structure and Formulation

The Hybrid MOABC–NSGA-II algorithm combines the global exploration capabilities of the Artificial Bee Colony (ABC) with the elitist sorting and diversity preservation features of NSGA-II, thereby effectively addressing constrained multi-objective optimization problems like CMMOPF. [13]In the ABC phase, each candidate solution is represented as $X_i = \{x_{i1}, x_{i2}, \ldots, x_{iD}\}$, where D is the dimensionality and $i = 1, \ldots, N$. Initial solutions are generated as:

$$x_{ij} = x_j^{\min} + \text{rand}[0, 1] \cdot (x_j^{\max} - x_j^{\min}). \tag{11}$$

New solutions are produced via perturbations in the employed and onlooker bee phases:

$$x'_{ij} = x_{ij} + w \cdot \text{rand}[-1, 1] \cdot (x_{ij} - x_{kj}), \tag{12}$$

where w is a weight factor, and $k \neq i$ is a randomly selected index. The selection probability is proportional to solution fitness, computed as:

$$\text{fit}_i = \frac{\#\text{ solutions dominated by } x_i}{N}, \quad P_i = \frac{\text{fit}_i}{\sum_{j=1}^{N} \text{fit}_j}. \tag{13}$$

Stagnant solutions are substituted through a scout mechanism upon exceeding a specified trial limit.

During the NSGA-II phase, offspring are generated using simulated binary crossover and polynomial mutation operators, which are subsequently subjected to fast non-dominated sorting and crowding distance computation to ensure diversity. [13]

$$\text{CD}(i) = \sum_{m=1}^{M} \frac{f_m(i+1) - f_m(i-1)}{f_m^{\max} - f_m^{\min}}. \tag{14}$$

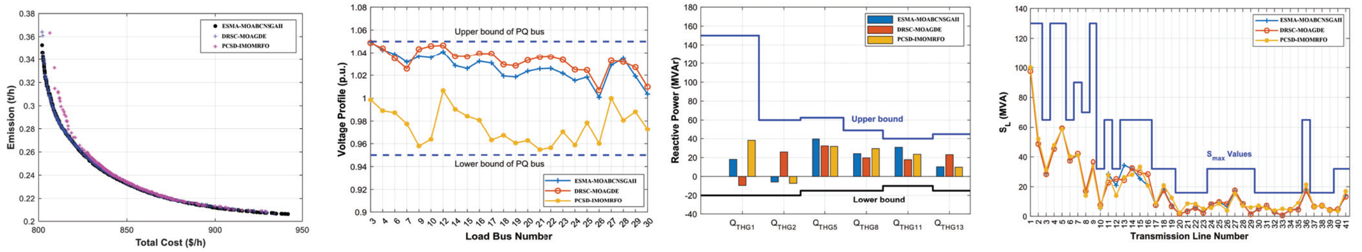

Fig. 2. For Case 1 : (a)Pareto Front curve (b) voltage profiles of the load buses (c) reactive powers of generating units (d) apparent power of the transmission

Selection is governed by a crowded-comparison operator ϕ_n:

$$x_i \phi_n x_j \iff (r_i < r_j) \vee [(r_i = r_j) \wedge (\mathrm{CD}_i > \mathrm{CD}_j)], \quad (15)$$

where r_i denotes the non-domination rank of individual i.

The hybridization involves alternating between ABC-based solution generation and NSGA-II-based selection. Constraint handling emphasizes the identification of feasible solutions, with infeasible individuals assessed based on the extent of their violations.

$$\text{rank}_{\text{infeasible}} = \text{rank}_{\text{max feasible}} + \text{sorted index by violation}. \quad (16)$$

This dual-phase synergy enhances convergence, preserves decision space diversity, and ensures robust constraint satisfaction, making it suitable for complex bi- and tri-objective CMMOPF scenarios.

Algorithm 1 Hybrid MOABC–NSGA-II with ESMA and Penalty-Based Constraint Handling

Input: $\{f_1, \ldots, f_m\}$, system data, ESMA model $\hat{f}$, penalties λ_e, λ_i
Output: Pareto-optimal archive $\mathcal{A}^*$
Initialize population $X = \{x_i\}_{i=1}^N$, $tc_i \leftarrow 0$
Evaluate x_i via Newton–Raphson; compute $\tilde{f}(x) = f(x) + \lambda_e \sum |h_k(x)| + \lambda_i \sum \max(0, g_j(x))$
Train ESMA on $\{x_i, \tilde{f}(x_i)\}$
for $g = 1$ to $G_{\max}$ **do**
  for each employed bee x_i **do**
    Generate x_i'; estimate $\hat{f}(x_i')$ via ESMA
    if x_i' dominates x_i **then** $x_i \leftarrow x_i'$, $tc_i \leftarrow 0$ **else** $tc_i \leftarrow tc_i + 1$
  end for
  Compute fitness $\tilde{f}$ and selection prob. P_i
  for each onlooker bee **do**
    Select x_i via P_i; generate x_i'
    if x_i' dominates x_i **then** $x_i \leftarrow x_i'$, $tc_i \leftarrow 0$ **else** $tc_i \leftarrow tc_i + 1$
  end for
  for x_i with $tc_i >$ limit **do**
    Replace x_i; reset tc_i
  end for
  Generate offspring via SBX (p_c) and mutation (p_m)
  Evaluate offspring via surrogate or exact functions
  Apply fast non-dominated sorting and crowding distance
  Select top N for next generation
  Update archive $\mathcal{A}$ with non-dominated solutions
  if $g \mod T_{\text{update}} = 0$ **then**
    Retrain ESMA on $\mathcal{A}$
  end if
end for
return Final archive $\mathcal{A}$

The MOABC–NSGA-II algorithm addresses the CMMOPF problem by combining the global exploration strengths of ABC with the elitist sorting and diversity preservation of NSGA-II. 12 ESMA, incorporating RBF, Kriging, and SVR, enhances convergence by approximating objective functions and cutting computational costs. Constraints are handled using a penalty method with Newton-Raphson for equality constraints. The algorithm iteratively updates solutions using ABC and NSGA-II phases, with periodic retraining of the surrogate model and dynamic archive updates to maintain convergence quality and Pareto diversity.

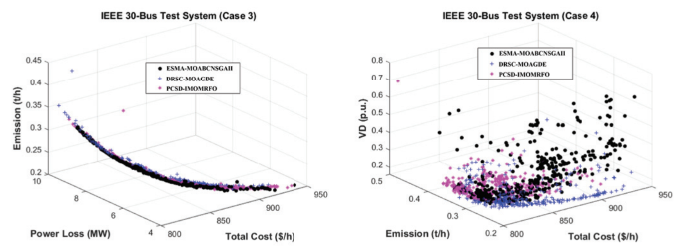

Fig. 3. Pareto Front Curve for Case 3 and 4

V. Test System and Case Descriptions

The ESMA–MOABC–NSGA-II framework was validated with the IEEE 30-bus test system, featuring 6 generators, 41 lines, 4 transformers, and multiple load and shunt buses. Load flow calculations used MATPOWER 8.0 and the Newton-Raphson method in MATLAB R2022b. [13] The framework was tested in six multi-objective scenarios (Cases 1–6), each with various combinations of fuel cost, emissions, voltage deviation, and power loss. Case 5 showcased the most comprehensive scenario, meeting all objectives at once.

VI. Simulation Results and Discussion

The proposed ESMA-MOABCNSGAII framework's effectiveness was assessed through extensive simulations on the IEEE 30-bus test system across six multi-objective scenarios, as shown in Table II. ESMA-MOABCNSGAII's performance was assessed against competitive methods found in the literature, such as DRSC-MOAGDE, IMOMRFO, and MOEAD-5. The proposed method showed better performance in most scenarios, with improved convergence, diversity, and compliance with constraints. Table 1 summarizes the results for all six cases.

The proposed ESMA-MOABCNSGAII achieved optimal solutions in all scenarios. In Case 1, fuel cost dropped to

TABLE III
OPTIMIZED CONTROL AND STATE VARIABLES FOR IEEE 30-BUS SYSTEM (CASES 1–6)

Bus	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
P_{THG1}	147.7332	178.7406	133.4466	164.8526	166.3839	153.6691
P_{THG2}	50.2548	48.4879	50.3252	56.2839	51.1255	50.3348
P_{THG5}	24.8216	21.7551	31.7749	24.2606	23.0117	25.1548
P_{THG8}	29.9347	19.9108	29.3174	18.7823	21.0714	25.2210
P_{THG11}	19.0975	18.0115	25.0175	11.6409	15.6541	13.4264
P_{THG13}	18.5196	12.4166	20.0015	16.7759	15.1601	22.6275
V_{THG1}	1.0792	1.0775	1.0470	1.0563	1.0492	1.0260
V_{THG2}	1.0592	1.0592	1.0354	1.0457	1.0524	1.1018
V_{THG5}	1.0251	1.0179	1.0296	0.9881	1.0030	1.0036
V_{THG8}	1.0365	1.0292	1.0105	1.0125	0.9952	0.9931
V_{THG11}	1.0945	1.0448	1.0710	1.0251	1.0699	1.0417
V_{THG13}	1.0542	1.0383	1.0440	1.0142	1.0433	1.0849
T_{11}	1.0492	0.9524	0.9411	0.9728	1.0516	1.0219
T_{12}	0.9775	1.0011	0.9834	1.0384	0.9916	1.0194
T_{15}	1.0609	1.0113	1.0254	0.9956	0.9983	1.0092
T_{36}	0.9927	0.9689	0.9344	1.0127	0.9814	0.9732
Q_{C10}	3.3506	2.2357	1.8939	3.7629	0.3576	1.3062
Q_{C12}	0.8200	3.7667	2.3760	1.4334	4.0720	3.4870
Q_{C15}	3.8895	4.4007	4.1077	4.9538	4.4603	4.6664
Q_{C17}	1.3329	2.5319	3.7326	1.2828	2.4533	2.2340
Q_{C20}	4.7713	3.3140	3.7243	4.3683	1.4047	2.1273
Q_{C21}	4.8550	1.2532	2.9061	3.2485	3.4815	3.4894
Q_{C23}	4.0253	4.4969	2.7094	3.3218	3.7643	3.9746
Q_{C24}	4.7815	1.1912	2.9722	4.8400	3.5792	3.6459
Q_{C29}	2.6555	2.8084	0.6670	2.2627	2.1173	2.1015
Q_{THG1}	18.0659	-5.3041	-10.2290	15.8686	-19.4473	12.9651
Q_{THG2}	-5.8255	26.9718	22.7312	38.7295	17.6421	26.9494
Q_{THG5}	25.7780	38.1267	48.1551	23.2351	41.6465	31.8787
Q_{THG8}	25.4647	27.2973	41.1516	25.4145	19.4715	20.3275
Q_{THG11}	30.8611	17.4390	14.0302	-3.1622	21.3821	20.8318
Q_{THG13}	17.2035	20.4978	16.6451	6.6129	14.2927	35.8823
TFC	**807.1634**	**801.0090**	**822.0900**	**805.1578**	**804.1732**	**810.1133**
TE	**0.2805**	**0.3708**	**0.2488**	**0.3327**	**0.3368**	**0.3022**
P_{Loss}	7.1235	3.9268	6.4431	2.1991	8.9067	8.2238
VD	0.6484	0.2719	0.3469	0.2329	0.2018	0.2124

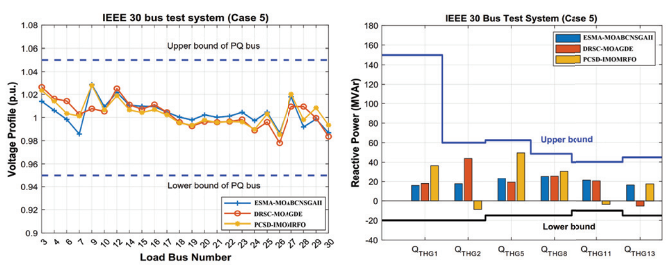

Fig. 4. Case 5: (a) voltage profiles of the load buses, (b) reactive powers of generating units

$808.96 per hour, leading to emissions of 0.2955 tons per hour, as illustrated in Fig. Fig. 2. The Pareto-front curves for Cases 3 and 4 are shown in Fig. Fig. 3. Case 5 achieved balanced optimization by considering all four objectives, resulting in a reduced voltage deviation of 0.1934 p.u. and a power loss of 3.98 MW, outperforming existing methods. The ensemble metamodel accurately predicted viable solutions using RBF, Kriging, and SVR, greatly reducing computational demands.

VII. PERFORMANCE ASSESSMENT AND VALIDATION

The proposed ESMA–MOABC–NSGA-II framework demonstrates superior performance compared to benchmark algorithms such as DRSC–MOAGDE and PCSD–IMOMRFO across all six constrained multimodal MOOPF scenarios on the IEEE 30-bus system as shown in Fig. 4 and Fig. 5. From Table III depicts DRSC–MOAGDE, the framework

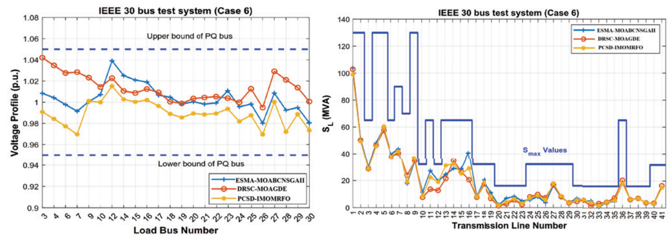

Fig. 5. Case 5: (a) voltage profiles of the load buses, (b) Apperant Power Flow in transmission

demonstrates average reductions of 0.86% in total fuel cost, 4.31% in emission levels, 30.29% in voltage deviation (VD), and 49.19% in active power loss. From Table IV presents PCSD–IMOMRFO, it demonstrates enhancements of 1.92% in cost, 6.18% in emissions, 55.78% in vehicle delay (VD), and 44.88% in power loss. The results indicate enhanced convergence, preservation of diversity, and effective constraint-handling in complex MOOPF scenarios.

ESMA–MOABC–NSGA-II shows significantly lower runtime and complexity than DRSC–MOAGDE and PCSD–IMOMRFO regarding computational efficiency. The improvement comes from using the ESMA, which cuts down power flow evaluations by periodically retraining surrogates. The DRSC–MOAGDE method uses costly global differential evolution, while PCSD–IMOMRFO struggles with memory-heavy recombination and sorting. The proposed method balances exploration and exploitation while reducing computational demands. Results show an average success rate of 88.69% and a runtime of 3.01 s, outperforming DRSC–MOAGDE (9.47 s) and PCSD–IMOMRFO (1372.49 s), as in Table V highlighting its strong applicability for real-time power system use.

The algorithm's computational complexity comes from three main components: metaheuristic search, surrogate modeling, and constraint evaluations based on power flow. The complexities of MOABC and NSGA-II are $\mathcal{O}(G_{\max} \cdot N \cdot D)$ and $\mathcal{O}(G_{\max} \cdot N \cdot \log N)$, respectively. ESMA surrogate modeling incurs retraining costs of $\mathcal{O}(N^3)$ for Kriging and $\mathcal{O}(N^2 D)$ for RBF and SVR, spread across planned update cycles. The proposed framework shows better scalability and lower computational overhead compared to DRSC–MOAGDE ($\mathcal{O}(G_{\max} \cdot N \cdot D)$ without surrogate) and PCSD–IMOMRFO ($\mathcal{O}(G_{\max} \cdot N^2)$ due to memory-based dominance sorting), making it more efficient for constrained multimodal optimization in power systems.

VIII. CONCLUSION

This study proposes ESMA-MOABCNSGAII, a hybrid framework for efficiently addressing CMMOPF challenges. It integrates the global exploration ability of MOABC with the elitist sorting and diversity preservation of NSGA-II, ensuring strong convergence and solution diversity. The Ensemble Surrogate Metamodel Archive, built with RBF, Kriging, and SVR,

TABLE IV

COMPARATIVE PERFORMANCE OF DIFFERENT ALGORITHMS ON IEEE 30-BUS SYSTEM [1], [6]

Methods	**Case 1**		**Case 2**		**Case 3**		
	Cost ($/h)	**Emission (t/h)**	**Cost ($/h)**	**VD (p.u.)**	**Cost ($/h)**	**Emission (t/h)**	**Power Losses (MW)**
ESMA–MOABC–NSGA-II	**807.1634**	0.2805	**801.0090**	0.2719	**822.0900**	0.2488	6.4431
DRSC–MOAGDE	808.9646	0.2955	801.2070	0.2733	824.0958	0.2688	6.4830
PCSD–IMOMRFO	817.9615	0.2736	801.3908	0.4435	855.0115	0.2318	5.2886
MOEA/D–SF	829.5150	0.2501	802.4060	0.1362	881.0120	0.2164	4.1441
MOEA/D	833.7200	0.2438	799.9900^*	0.3540	902.5400	0.2107	**3.4594**
MOPSO	833.8600	0.2483	800.0300^*	0.4422	891.4800	0.2144	3.9557
NSGA-II	835.5900	0.2449	800.0600^*	0.4486	903.7900	**0.2103**	3.7917
MOMICA	865.0660	**0.2221**	804.9600	**0.0952**	–	–	–
MOICA	865.3184	0.2246	805.0345	0.1004	–	–	–
MPIO–PFM	833.1703	0.2397	–	–	866.0601	0.2160	4.4474
MPIO–COSR	832.4655	0.2351	–	–	863.9503	0.2126	4.3177

TABLE V

COMPARATIVE PERFORMANCE OF DIFFERENT ALGORITHMS ON IEEE 30-BUS SYSTEM (CASES 4–6) [1], [6]

Methods	**Case 4**			**Case 5**			**Case 6**			
	Cost ($/h)	**Emission (t/h)**	**VD (p.u.)**	**Cost ($/h)**	**VD (p.u.)**	**Power Losses (MW)**	**Cost ($/h)**	**Emission (t/h)**	**VD (p.u.)**	**Power Losses (MW)**
ESMA-MOABCNSGAII	**805.1578**	0.3327	0.2329	**804.1732**	0.2018	8.9067	**810.1123**	0.3022	0.2124	8.2238
DRSC-MOAGDE	806.2563	0.3348	0.2465	804.4002	0.2022	9.0067	810.7423	0.3069	0.2283	8.3876
PCSD-IMOMRFO	811.8350	0.2908	0.2206	806.7303	0.3720	7.8498	816.4599	0.2785	0.2738	7.2697
MOEA/D-SF	842.446	0.2406	**0.1092**	836.711	**0.1258**	5.8223	883.322	**0.2187**	**0.1322**	**4.4527**
MOEA/D	850.28	**0.2332**	0.1155	831.81	0.1355	5.9926	-	-	-	-
MOPSO	846.93	0.2386	0.2188	827.82	0.1588	6.5929	-	-	-	-
NSGA-II	825.86	0.2648	0.1421	843.14	0.1931	6.4917	-	-	-	-
B-MMOFPA	-	-	-	843.18	-	**5.7886**	-	-	-	-
MOMICA	-	-	-	-	-	-	830.188	0.2523	0.2978	5.585

effectively estimates fitness landscapes and reduces computational effort. Dynamic Reference Space clustering and Pareto-Crowded Sorting Dominance further enhance archive maintenance and multimodal exploration. Tests on the IEEE 30-bus system show superior performance over DRSC-MOAGDE and PCSD-IMOMRFO, achieving 88.69% success, 3.01 s computation, and optimal trade-offs in fuel cost, emissions, voltage deviation, and power loss. By combining decomposition-based cooperative search with surrogate-assisted learning, the framework offers scalability and robustness for high-dimensional, constrained problems. Future work will extend this approach to dynamic OPF with uncertainty modeling and real-time surrogate retraining.

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